

PANDA Component Description: *fifo_8bit*

Overview

The *fifo_8bit* component implements a number of simple 8-Bit FIFOs. Number and size of the FIFOs can be changed through generics.

The *fifo_8bit* component is designed to use a minimal number of memory-blocks.

Generics

Name	Type	Default	Description
num_of_fifos	integer	3	$2^{\text{num_of_fifos}}$ FIFOs are implemented (1...8)
size_of_fifos	integer	64	Number of bytes per FIFO

Registers

Data Register

Name: DAT
Address: Base + 0x00

Name	DAT
R/W	R/W
Reset Value	0x00

A write operation on *DAT* pushes the written data into the selected FIFO. A read operation on *DAT* pops data from the selected FIFO. The FIFO is selected with the *SEL* register.

Select Register

Name: SEL
Address: Base + 0x01

Name	SEL
R/W	R/W
Reset Value	0x00

The FIFO that is used can be selected through the *SEL* register. The width of *SEL* is specified in *num_of_fifos*. *SEL* is right adjusted and unused bits are read as '0'.

Status Register

Name: STATUS

Address: Base + 0x02

Name	-	-	-	-	-	-	EMPTY	FULL
R/W	-	-	-	-	-	-	R	R
Reset Value	0	0	0	0	0	0	1	0

FULL: '1' → The selected FIFO is full
 '0' → The selected FIFO is not full

EMPTY: '1' → The selected FIFO is empty
 '0' → The selected FIFO is not empty